Tokenized Equity Trading on Decentralized Exchanges

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CBER Crafting the Cryptoeconomy, Columbia, Oct 30 2026

What this talk covers

1

How compliance gets built onchain

Where controls can be enforced: Robinhood Chain's chain-level filtering, Permissioned Pools at the contract level, app-level compliance, and the SEC's new exemption.

2

What the market shows

When people trade tokenized stocks, what those prices mean, and two weekends that tell very different stories.

3

What to build next

The caveats we ran into, and ideas worth building around them.

In traditional finance, intermediaries enforce the rules

Investor → **Broker**
Checks who you are: KYC, anti-money-laundering, eligibility → **Exchange**
Controls what trades and how: listings, halts, surveillance → **Transfer agent**
Records who owns it and enforces transfer restrictions

> DeFi removes these intermediaries. So where do the enforcement points go?

Park (2026), "DeFi vs. TradFi: Institutions and Industrial Organization," p. 3.

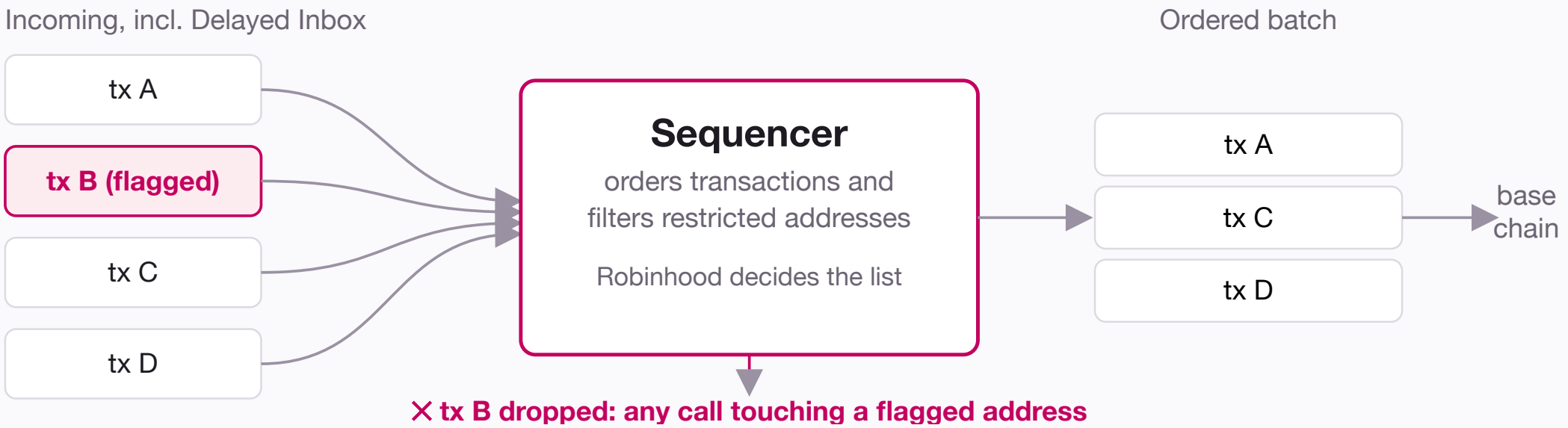
Where can compliance live?

Infra	**The chain** decides who can transact	L2s and single-sequencer chains
Code	**The token and pool contracts** decide who can hold or trade	Token transfer rules, Permissioned Pools
App	**The frontend or broker** decides who can interact with the code	KYC at onboarding, venues

Permissioning can sit at the infrastructure, platform or asset level: IMF GFSR (Oct 2026), Table 3.1.

PART 1 · INFRA

Robinhood Chain: compliance at the sequencer



LIVE SINCE

July 2026

Filtering at the chain owner's discretion

AFTER MINT

77%

Uniswap's share of stock-token DEX volume, in open pools against USDG

TRADE-OFF

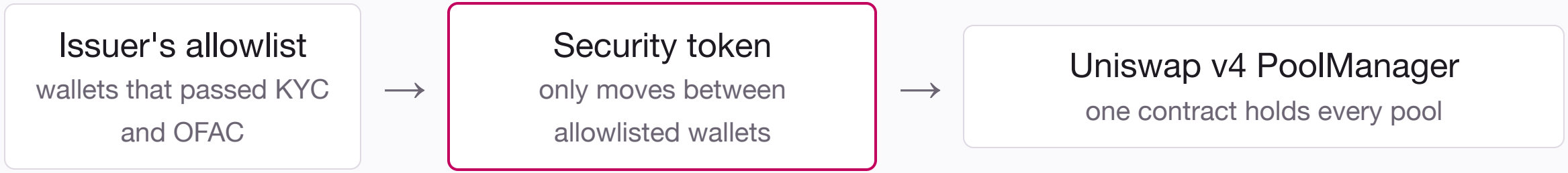
Operator holds the keys

Less centralization means compliance has to live in the contracts

Uniswap Labs data, Jul 26 to Oct 23, 2026, wash trading excluded. All data in Part 2 comes from these pools.

PART 1 · CODE

Token transfer restrictions, and the shared-pool problem



- Tokens can enforce their own rules in code: a security token checks its allowlist on every transfer.
- A whitelist has to include every contract the token touches.
- To sit in a v4 pool, the token would have to allowlist the PoolManager, which means every pool at once.

The whitelist problem: Malinova and Park (2023), p. 15.

PART 1 · CODE

Hooks: why a security token needs one more layer

A v4 pool is defined by

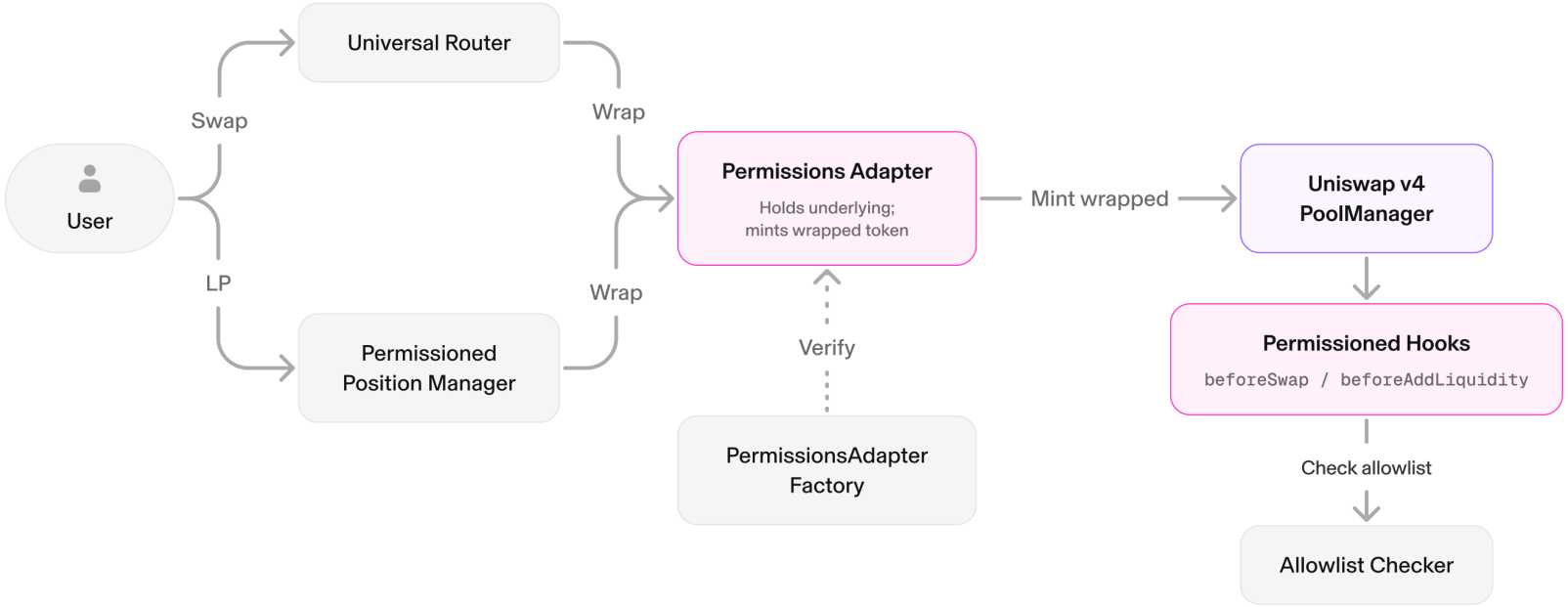
1	Token 0
2	Token 1
3	Fee
4	Tick spacing
5	**Hook**

- A hook is a contract that runs before and after events: initializing a pool, swapping, adding or removing liquidity.
- If the hook reverts, the trade doesn't happen.
- That makes it a natural place to enforce trading permissions at the pool level.

Bachu, Hasbrouck, Saleh and Wan (2026), "An Overview of Uniswap v4"; their CBER 2025 talk is recorded.

PART 1 · CODE

Permissioned Pools: the architecture



Wrapper: holds the real NVIDIA tokens and puts a wrapped copy, one for one, into Uniswap. The issuer approves only this one contract.

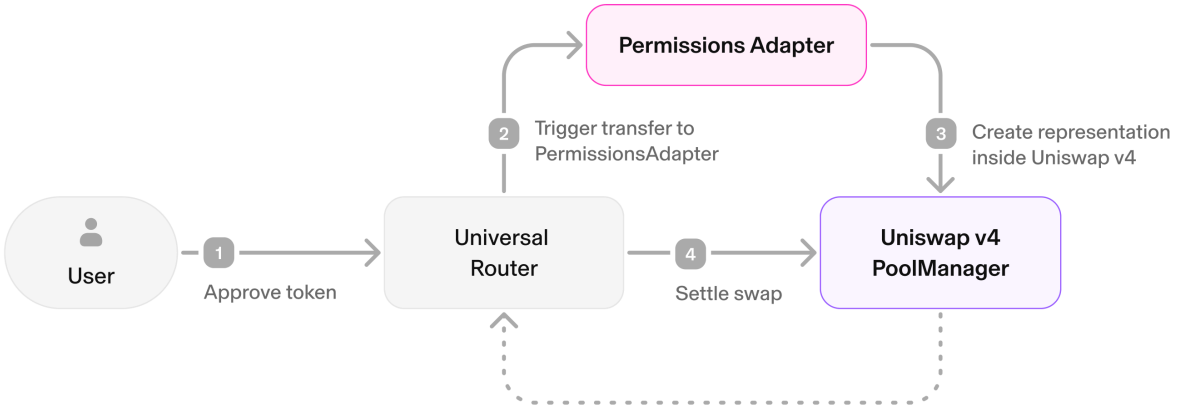
Issuer picks: which routers (the contract your wallet calls to trade) and which hooks can touch the wrapped copy. Trading and providing liquidity are approved separately.

Proof: an onchain allowlist, or a signed approval sent with each trade. The pool doesn't care which.

Diagram: developers.uniswap.org.

PART 1 · CODE

A swap through a permissioned pool



1
You want NVIDIA. Your wallet sends USDC to the **router**

2
The **hook** checks: is this wallet allowed to trade NVIDIA?

3
The pool swaps USDC for wrapped NVIDIA

4
The router unwraps it. You get NVIDIA.

If step 2 says no, the whole swap reverts. Same pool math as any other v4 pool.

Diagram: developers.uniswap.org.

PART 1 · APP

The SEC's exemption, and who's building on it

> *A Tokenized Securities Venue provides AMM pools for permissioned participants and sets standards for who can access trading.*
>
> Paraphrasing SEC Release 34-106402, Sept 2026, pp. 1, 7

THE PATH

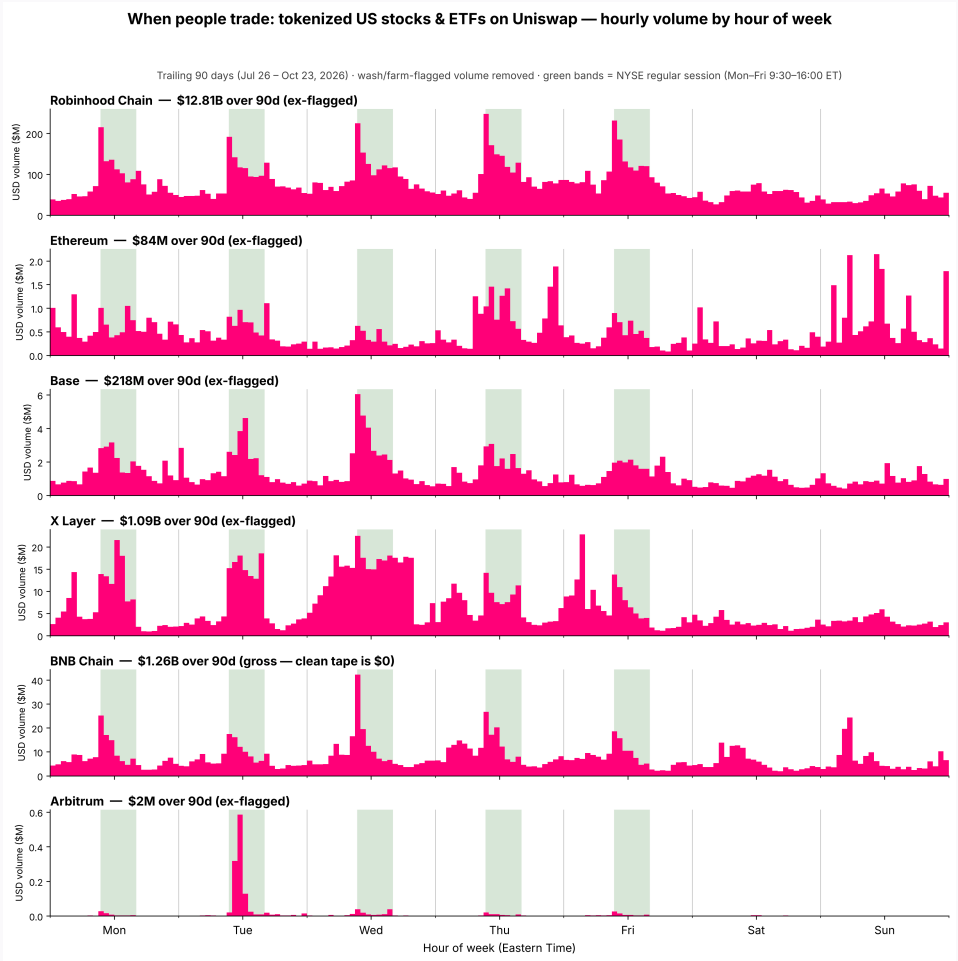
A venue operates permissioned AMM pools for tokenized stock, under conditions set by the SEC.

WHO'S BUILDING ON IT

OKXICE (notice, Oct 4): Uniswap v4 pools on X Layer, with its own hook built on top of Permissioned Pools.

PART 2 · WHAT THE MARKET SHOWS

Most trading happens while the market is closed



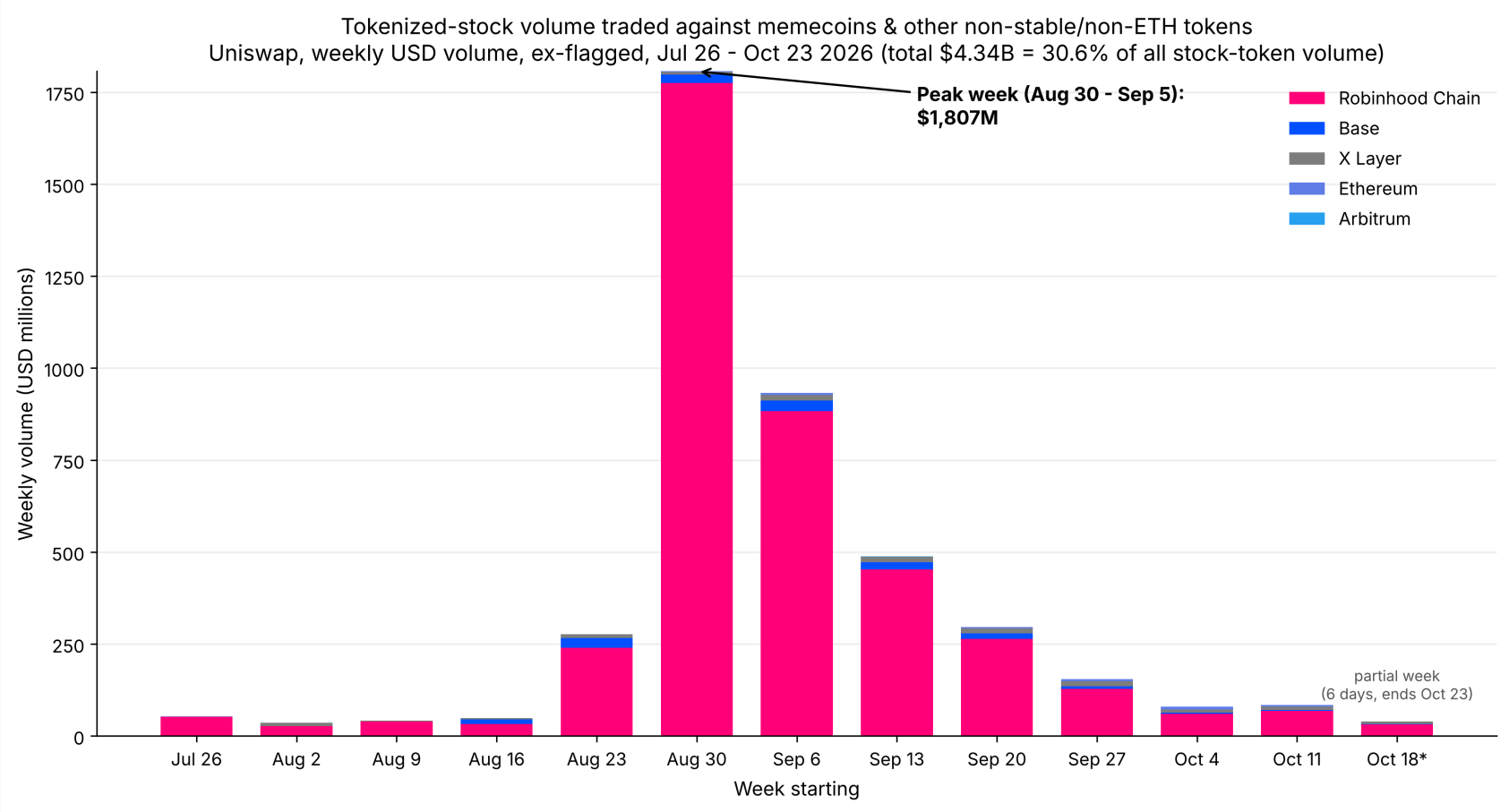
Chain	90-day volume	Off-hours
Robinhood	$12.8B	**66%**
X Layer	$1.1B	**61%**
Base	$218M	**64%**
Ethereum	$84M	**74%**
BNB	$0 after flags	farming

Small, retail, on a US clock. Median off-hours trade $48; 74% are under one share; the top 10 wallets are under 5% of Robinhood Chain's off-hours volume. Volume follows US sessions, not European working hours.

Uniswap pools, tokenized US stocks and ETFs, Jul 26 to Oct 23, 2026: $14.2B after flags. Flags remove volume that looks like wash trading or points farming (16.5%), including all of BNB.

PART 2 · WHAT THE MARKET SHOWS

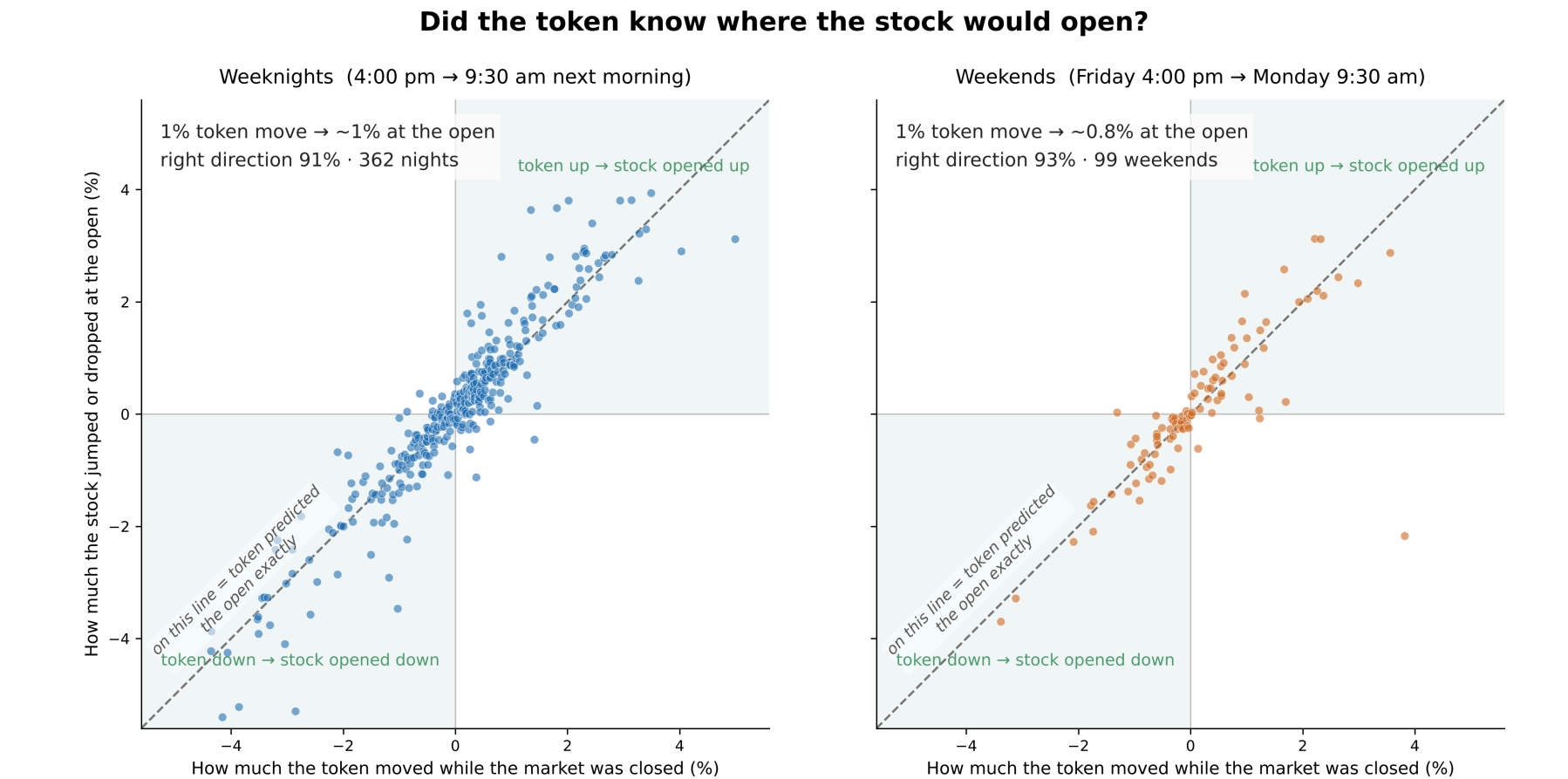
30% of the volume traded against memecoins



Pools where a stock token is the quote asset for a memecoin, e.g. "AI" priced in NVIDIA, "MOO" in Micron, "FATCOIN" in Eli Lilly. $4.3B of $14.2B, 93% on Robinhood Chain. Uniswap pools, ex-flagged, Jul 26 to Oct 23, 2026.

PART 2 · WHAT THE MARKET SHOWS

The token's price tells you where the stock will open

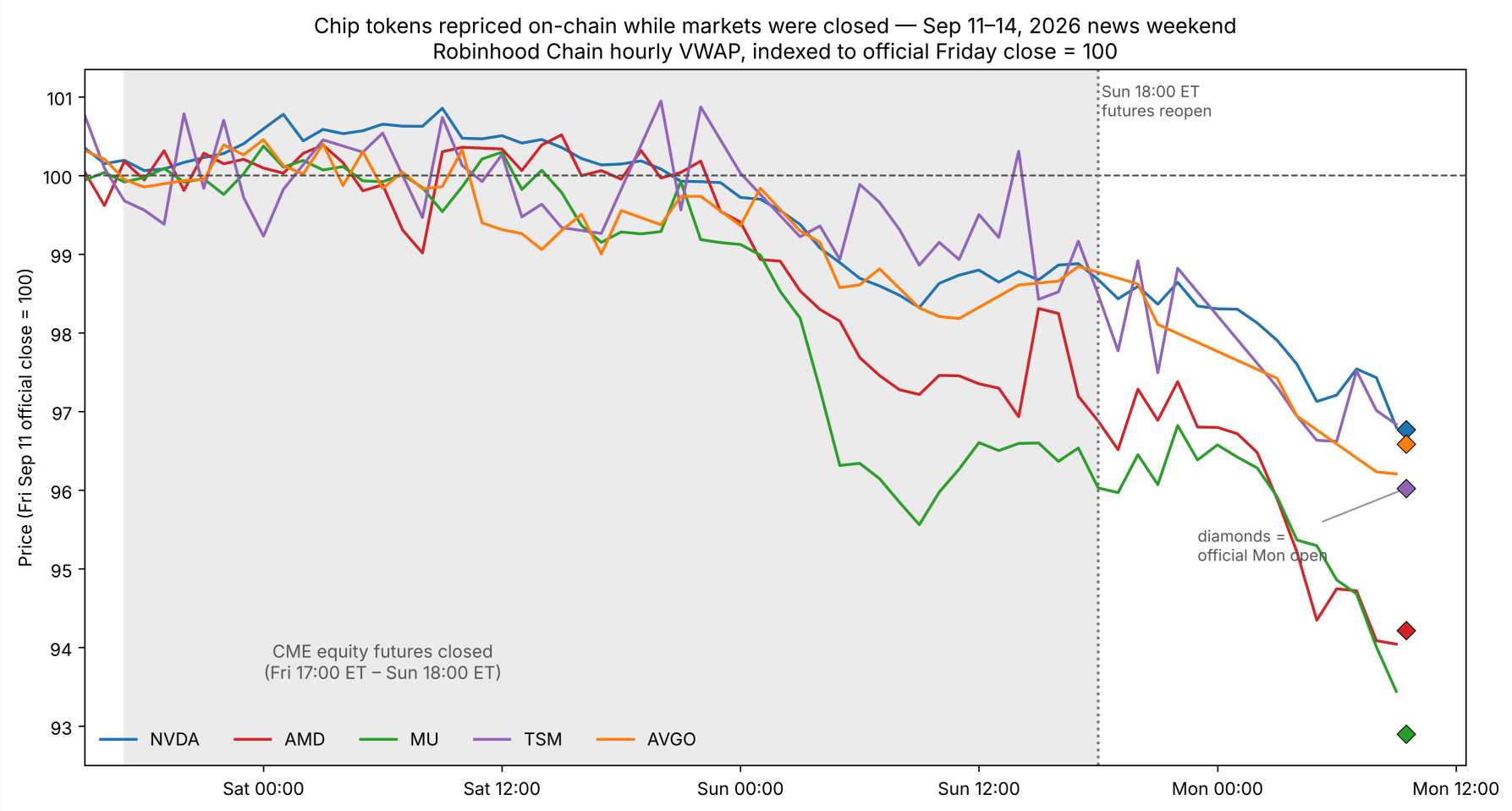


Robinhood Chain tokenized stocks (fake-print-cleaned) vs the underlying stock's official opening gap, Jul 10 – Oct 23 2026. Token moves measured with 60-min VWAPs at the window boundaries; overnight moves larger than 5% excluded. Official OHLC reference ends at the Oct 13 open, so plotted windows run Jul 14 → Oct 13 (weeknights) and Jul 13 → Oct 12 (weekends).

Each dot is one stock on one night or weekend. Robinhood Chain, Jul to Oct 2026. Agrees with Cong et al. on aggregated token prices and the IMF's 87-99% on CEX prices.

PART 2 · CASE STUDY

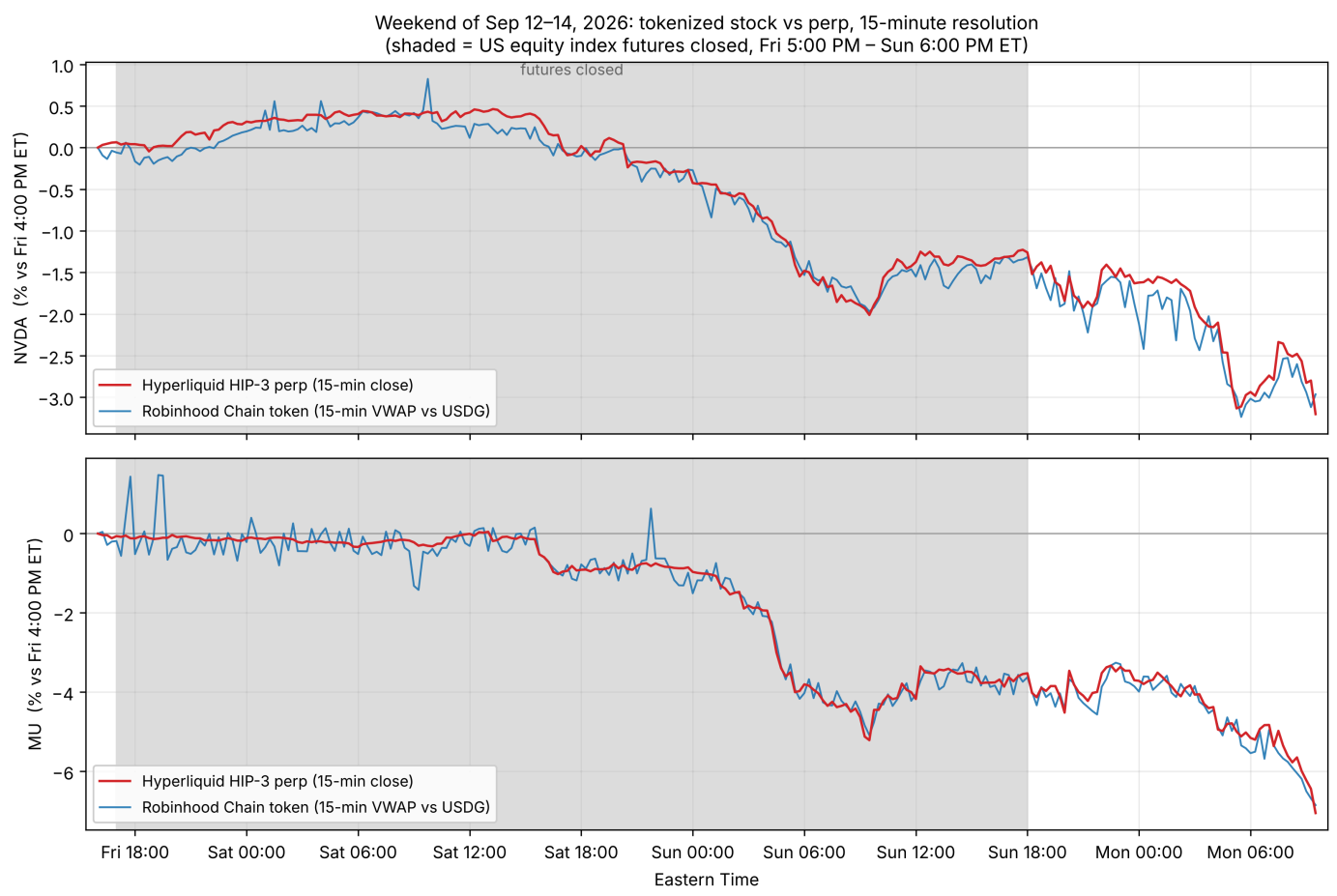
Sep 12-14: onchain markets priced the news first



Robinhood Chain hourly prices vs Friday's close; diamonds are Monday's official opens. No futures and no mint or redeem until Sunday evening. News: Anthropic's CEO called for slowing AI development.

PART 2 · CASE STUDY

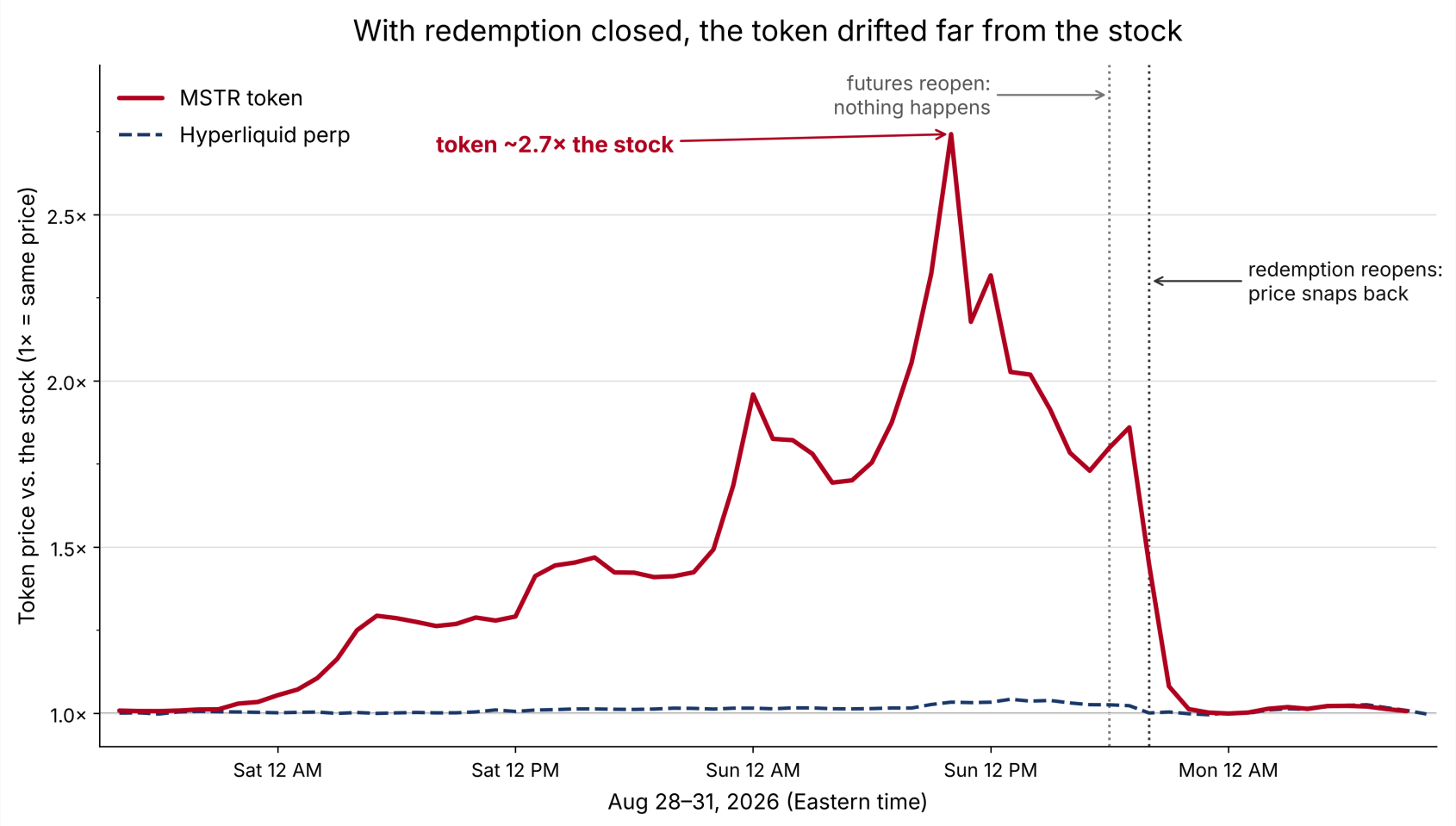
Same weekend: pools and perps moved together



Hyperliquid HIP-3 perps vs Robinhood Chain tokens, 15-minute intervals. Largest hourly gap between token and perp: NVDA 0.6%, MU 0.9%, AMD 1.3%.

PART 2 · CASE STUDY

Aug 28-31: the same pools can come unhinged



Robinhood never mints or redeems on Saturdays, or on Sundays before 8 PM ET. Premiums held through the 6 PM futures reopen and broke at 8 PM. Where a perp existed (MSTR, ASML, MRNA), it stayed near the stock while the token ran.

PART 3 · WHAT TO BUILD NEXT

Caveats we ran into, and ideas worth building

Caveat	Idea: a hook or a pool type
The same stock trades as several tokens, each with thin pools	**Same-stock pools across issuers**, priced like a stablecoin pair
Thin, newly listed tokens are fragile on their own	**Correlated pairs:** a stock against its ETF or same-industry names
Outside regular hours, brokers aren't held to the best national price (Huang and Schwarz)	**Oracle-priced pools:** pin execution to the national best bid and offer
Weekends have no redemption; a perp shows the fair price but can't enforce it	**Weekend hook:** band the pool around a 24/7 reference price while mint and redeem are shut

CLOSE

Wrapping up

PART 1

Compliance can live in the chain, the contracts or the app. All three exist today.

PART 2

Token prices call the open. On news weekends they move first. Without redemption, they can also come unhinged.

24/7 ACCESS	FRACTIONAL	COLLATERAL	COMPOSABLE

TWO OPEN QUESTIONS

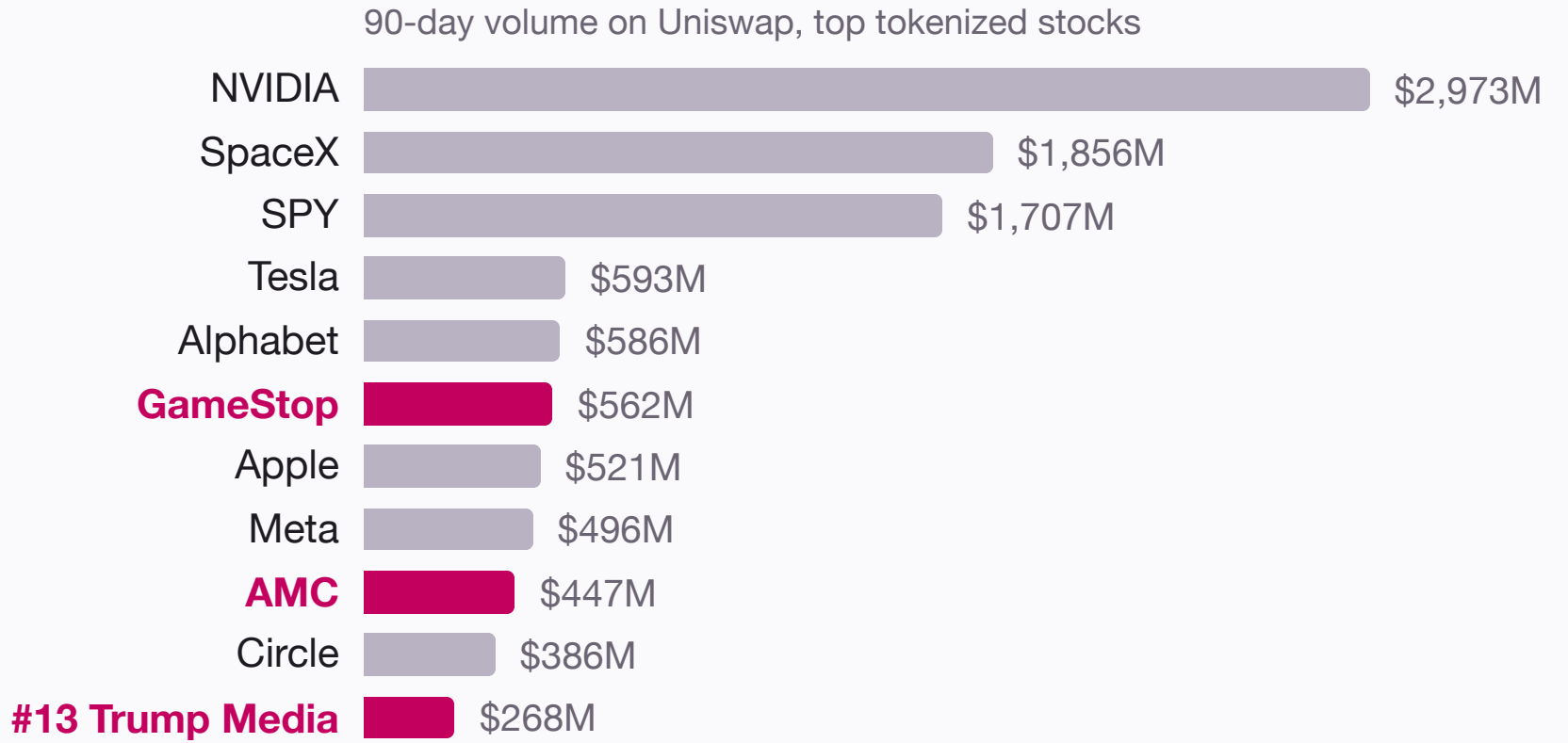
> Is redemption actually needed for price discovery, or just a deep enough source of price information? And if the information is already out there, what forces the pool to follow it?

> Onchain, a stock can be paired with anything: a memecoin, its own ETF, another company's token of the same stock. Which of these pairings are new markets worth building?

Backup

For Q&A

Meme stocks trade like megacaps onchain



GameStop out-trades Apple.

GameStop + AMC ≈ Alphabet + Meta.

On Nasdaq, they trade a tiny fraction of what those megacaps do.

Uniswap pools, all chains, ex-flagged, Jul 26 to Oct 23, 2026. Circle and Trump Media are approximate. About 99% of meme-stock token volume is on Robinhood Chain.

Who controls what in a permissioned pool

Control	Who holds it
Who can trade (Allowlist Checker)	The issuer, usually as an onchain whitelist
Which routers and hooks may wrap the token	The issuer, through its Permissions Adapter
Venue-specific logic	A venue's own hook on top of Permissioned Pools, e.g. OKXICE
Who counts as a venue under the exemption	The order's definition; for any company, a question for its lawyers

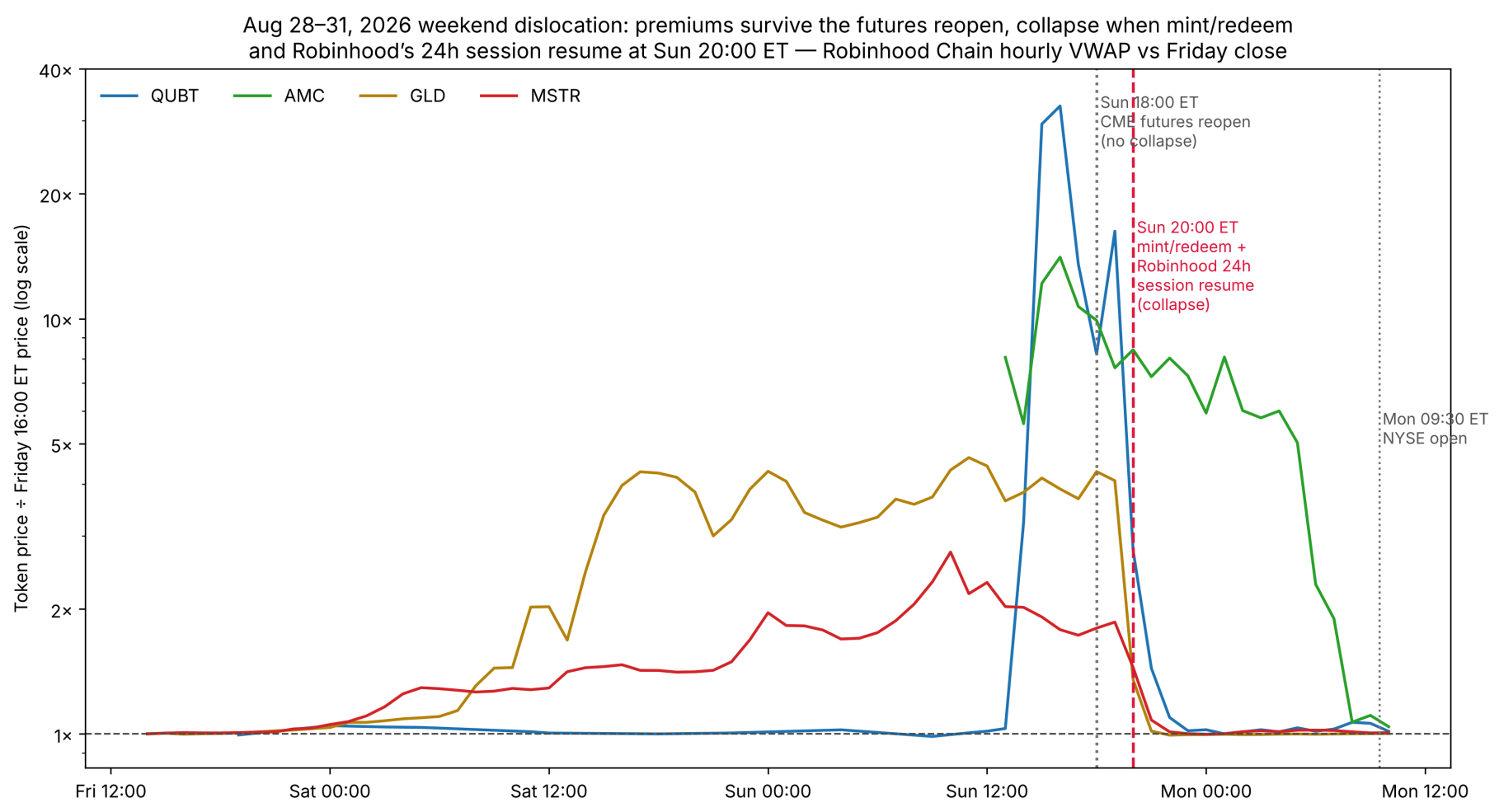
SEC Release 34-106402, pp. 1, 7, 9-10.

The weekend numbers

Window	Share of the token's move that shows up at the open
Weeknights, 4 PM to 9:30 AM	1.05, right direction 91%, 362 nights
Whole weekend, Friday close to Monday 9:30	0.83, right direction 93%, 99 weekends
Quiet weekends, Friday 5 PM to Sunday 6 PM only	about 0: the price drifts and comes back
Sep 12-14, Friday 5 PM to Sunday 6 PM	about 44% of Monday's gap priced in advance; correlation 0.93 across 14 stocks
Oct 16-18, news after futures reopened	tokens carried none of it

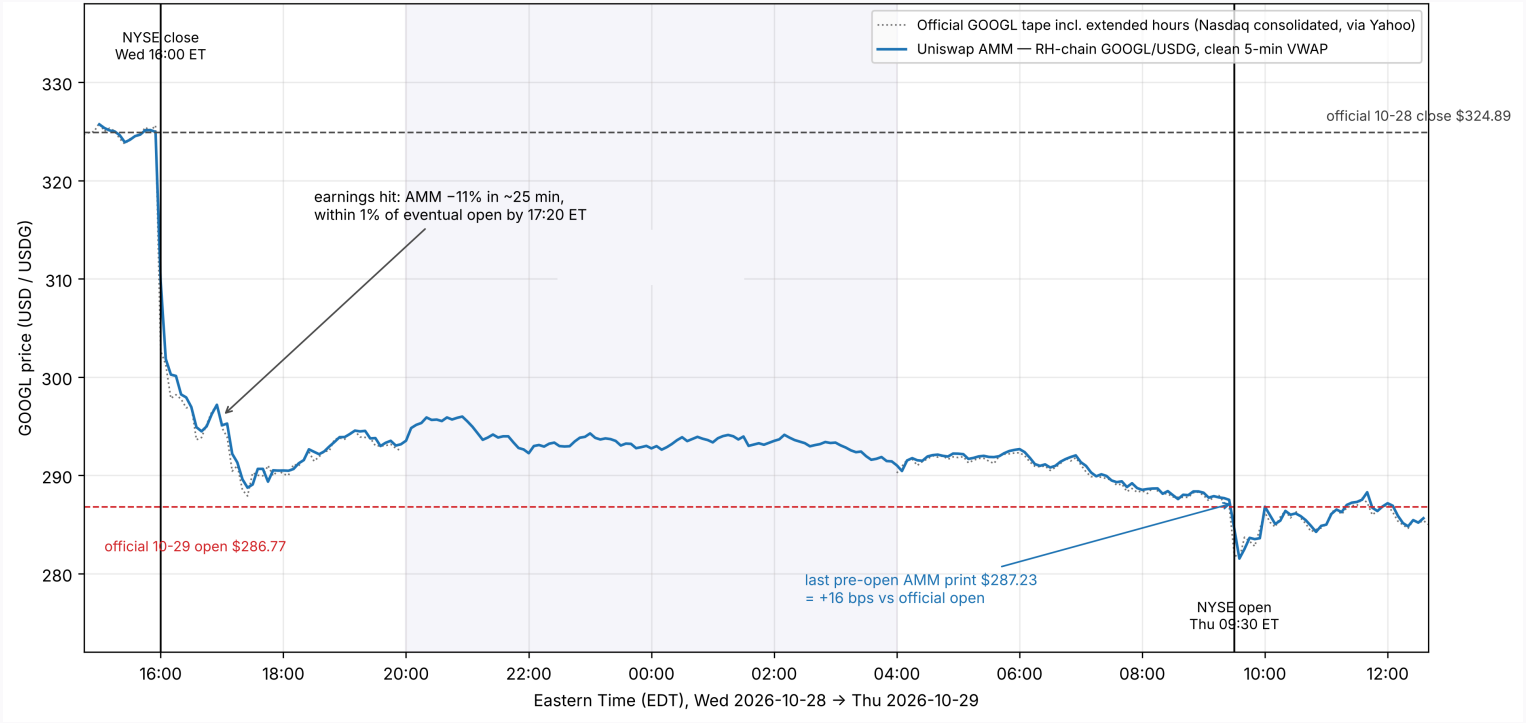
Robinhood Chain, Jul 10 to Oct 19, 2026; 60-minute average prices at each boundary; official opens and closes; moves above 5% trimmed. Futures reopen Sunday 6 PM ET.

Aug 28-31: premiums broke when redemption reopened



Robinhood Chain hourly prices vs Friday's close, log scale. Futures reopened at 6 PM Sunday; mint/redeem and Robinhood's 24-hour market at 8 PM.

Google earnings, Oct 28: the token tracked after hours



AFTER HOURS
About 85% of the 11.7% drop, mostly within 25 minutes

OVERNIGHT
Flat, around $293

BEFORE THE OPEN
5-minute average within 0.26% of the $286.77 open

Uniswap on Robinhood Chain, GOOGL/USDG, 5-minute average prices, wash trading excluded. Official prices incl. extended hours via Yahoo. Overnight venues not shown. Thin overnight tape: about $292K, 8 PM to 4 AM.

Liquidity providers off-hours

- Off-hours fills, marked to the next official open: adverse selection costs LPs about **28% of fees**; net **+5.5 bps**
- Weekend losses land when the reference markets reopen, not during the closed hours
- Closed NVDA/USDG 5 bp positions: **57.5%** beat holding, before gas

Uniswap Labs data, Oct 20.

MEV on Robinhood Chain

- One sequencer and no public mempool: nobody outside the sequencer sees a swap before it's ordered
- Sandwich-shaped patterns: about **0.3% of swaps**, and they lose money in aggregate
- Latency arbitrage: not yet measured

Uniswap Labs data, Oct 14.

The exemption's caps and conditions

- **Conditions that touch code:** permissioned access with OFAC screening; public contracts on a permissionless ledger; stop with a primary-market halt; USD trade feed within 10 minutes
- **Caps:** Tier 1, 75 symbols and 0.25% of consolidated volume per stock; Tier 2, 250 symbols and 2.5%
- Tokenized NVDA on Uniswap today is about 0.12-0.17% of NVDA's dollar volume
- Runs five years, to Sept 2031

SEC Release 34-106402, pp. 18, 24-30.